a) SST
b) profile clusters
c) clusters in T–S
d) T–S diagram
e) SST
f) profile clusters
g) clusters in T–S
h) T–S diagram
i) SST
j) profile clusters
k) clusters in T–S
l) T–S diagram
m) SST
n) profile clusters
o) clusters in T–S
p) T–S diagram
q) SST
r) profile clusters
s) clusters in T–S
t) T–S diagram
u) SST
v) profile clusters
w) clusters in T–S
x) T–S diagram

Data
No constraint
No constraint, border 0
TS constraint
Density constraint
Density monotonicity constraint

60°N 40°N 20°N 0° 20°S 40°S 60°S

T [°C]
S [g kg^{-1}]

SST [°C]
cluster, by increasing σ_0

k-means centre (data)
mean profile of the cluster

$\log_{10}$ volume fraction, ACC
$\log_{10}$ volume fraction, rest of the basin